

AE-530: Gas phase calibration of Normal range H2 sensor

Date: 2025-08-13

Tags: [Test](#) [python](#) [AE](#) [Calibration](#) [H2](#) [in-situ](#) [Unisense](#) [electrochemical H2 detection](#)

Category: H2 Evolution

Status: Done

Created by: Alexander Eith

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	H2 Evolution - AE-510: Gas phase test of H2 sensor H2 Evolution - AE-513: Gas and liquid phase calibration of H2 sensor

Reagents

Name	CAS Number / Experiment Number	Inventory number	pressure [bar]	Volume	mass [g]	Concentration [M]
hydrogen	1333-74-0	/	approx. 1	2 ballon	/	/
Milli-Q water	/	/	/	25 mL	/	/

Procedure/observations

Date	Time	Step	Observations	Pictures
13.80		The gas phase calibration according to Protocol - Gas phase calibration of H2 UniAmp sensor was started, with some deviations with Equipment - H2 UniAmp Sensor - Normal range - 2.1 x 80 mm needle . The steps taken are shortly presented below with timestamps and deviations and observations.	/	/
		The sensor was left connected to the device after H2 Evolution - AE-528: Initial test of H2 sensor (normal range) , therefore no conditioning was needed and therefore not done here. Also the flask containing H2 was used from AE-528		
	13:40	The setup with the H2/O2 reactor, the H2 sensor and the PT1000. The H2 sensor was connected using a 2 mm BOLA fitting. The reactor was filled with 25 mL water using a 20 mL syringe	The reactor was not totally dry after Prep work - AE-529: Volume determination of H2/O2 reactor	setup H2 sensor.jpg setup close up.jpg setup top view.jpg
	14:17	The H2 bubbling was resumed with a new ballon		

14:20	The logging was started with Pa as unit for the H2 content and a sample interval of 1 s	AE-530-1-logger1	/
14:21	The 0 ppm calibration point was taken	/	/
14:23	H2 (0.06 mL) was added into the liquid using a 1 mL syringe	As volume of the gasphase 56.82 mL was assumed according to Prep work - AE-529: Volume determination of H2/O2 reactor	/
14:26	The calibration point with 1056 ppm H2 was taken	Taken with intensity of 1.7 mV, noise level 0.25 mV	/
14:27	H2 (0.56 mL) was added into the liquid using a 1 mL syringe	/	/
14:27	The H2 bubbling was paused	/	/
14:29	The calibration point with 10912 ppm H2 was taken	Taken with intensity of 10.7 mV	/
14:30	H2 (1.0 mL) was added into the liquid using a 1 mL syringe	/	/
14:34	A 2nd ballon was added to the H2 flask	/	with 2nd ballon.jpg
14:35	The calibration point with 28511 ppm H2 was taken	Taken with intensity of 27.5 mV	
14:36	H2 (2 mL) was added into the liquid using a 2 mL syringe		
14:39	The calibration point with 63710 ppm H2 was taken	Taken with intensity of 58.0 mV	
14:39	H2 (3 mL) was added into the liquid using a 2 mL syringe		
14:43	The calibration point with 116508 ppm H2 was taken	Taken with intensity of 99.5 mV	
14:44	H2 (10 mL) was added into the liquid using a 20 mL syringe		
14:47	The calibration point with 292503 ppm H2 was taken	Taken with intensity of 209.7 mV	

	14:48	A hydrogen ballon was attatched to the reactor and H2 bubbling was started		cannula posiiton for H2 bubbling.jpg setup H2 bubbling in reactor.jpg
	14:51	The calibration point with 1.000.000 ppm H2 was taken	Taken with intensity of 1078 mV Noise similar to low range	
	14:52	The H2 bubbling of the reactor was stopped		
	14:53	The reactor was opened to the enviroment by opening the not used GL14 ports		
		The calbiration was saved	AE-530-calib1	/
		The calibration was saved without the 292503 ppm point	AE-530-calib2	
	15:16	The logging was stopped	/	/

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
13.08	14:20	AE-530-1-logger1	electrochemical H2 detection	Unisense H2 trace detector	water	AE-530-1.ulo g	AE-530-1-logger1.png AE-530-calib1.bmp AE-530-calib2.bmp	/	Worked well In total 8 point calibration was performed. 292503 ppm: to low signal: 2 possibiliteis: 1. error in amount H2 added (likely, due to use of 20 mL syringe), 2. error due to increased pressure, which was not shown in pressure sensor in device When taken all data points into account: slope: 0.011, intercept: -16.536, R2: 0.990 When ignoring datapoint at 292503 ppm: slope: 0.011, intercept: -6.389, R2: 1.000 Noise level cosistent at around 0.2 to 0.3 mV

Results

Worked very well, linear up to 100 %

Good response, approx. detection limit 500 ppm, 50 Pa

Linked experiments

H2 Evolution - [AE-492: Liquid phase calibration of H2 sensor](#)

H2 Evolution - [AE-498: Liquid phase calibration of H2 sensor](#)

H2 Evolution - [AE-507: Liquid phase calibration of H2 sensor](#)

H2 Evolution - [AE-508: Gas phase calibration of H2 sensor](#)

H2 Evolution - [AE-510: Gas phase test of H2 sensor](#)

H2 Evolution - [AE-513: Gas and liquid phase calibration of H2 sensor](#)

H2 Evolution - [AE-522: Gas phase calibration of H2 sensor I](#)

H2 Evolution - [AE-528: Initial test of H2 sensor \(normal range\)](#)

Prep work - [AE-477: Testing of in-situ electrochemical hydrogen sensor and test H2 evolution reaction with RhCrO:Al:SrTiO3](#)

Prep work - [AE-480: Testing of in-situ electrochemical H2 sensor from Unisense - gas phase and PT 100](#)

Prep work - [AE-529: Volume determination of H2/O2 reactor](#)

Linked resources

Equipment - [H2 UniAmp Sensor - Normal range - 2.1 x 80 mm needle - CEEC II E002 C2](#)

General Protocol - [Gas phase calibration of H2 UniAmp sensor](#)

General Protocol - [Liquid phase calibration of H2 UniAmp sensor](#)

Attached files

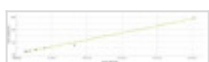
AE-530-calib2.bmp

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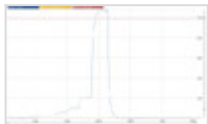
AE-530-calib1.bmp (Untitled)

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AE-530-1-logger1.png

sha256: b0c773a2d22c6bd8a553f3f76f52a501b132fb88522faf000871b9e5054e241a



with-2nd-ballon.jpg

sha256: aeefbb6f7591f3f384f11300334d3cd821fedd58764d2ee7a1e9a6a29629ba97



cannula-positon-for-H2-bubbling.jpg

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setup-H2-bubbling-in-reactor.jpg

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setup-top-view.jpg

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setup-close-up.jpg

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setup-H2-sensor.jpg

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AE-530-1.ulog

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Unique eLabID: 20250813-16ab10947e7eed03fb2e5fb8a31d4a2997e1c046

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=2673>